

Evidence-Gap Assessment: Software System Design Definition Assessment

Base Practice	Status	Rationale
Prepare for software system design definition	EvidenceStatus.PARTIAL_EVIDENCE	Claims suggest a lack of a formal architecture roadmap (CLAIM-001, CLAIM-004) necessary for defining a comprehensive design definition strategy. However, the selection of Apache Arrow based on specific criteria (CLAIM-002) supports prioritization of design principles and characteristics.
Establish designs related to each software system element	EvidenceStatus.PARTIAL_EVIDENCE	Apache Arrow's inclusion for in-memory data transfer and cross-language compatibility (CLAIM-002) shows transformation of architectural characteristics into design elements. However, the absence of formal trials (CLAIM-006) and a review process (CLAIM-007) indicates gaps in examining design alternatives and interface refinements.
Assess alternatives for obtaining software system elements	EvidenceStatus.PARTIAL_EVIDENCE	While Apache Arrow was chosen based on specific functionalities (CLAIM-002), the absence of formal trials for other protocols (CLAIM-006) indicates incomplete assessment of design alternatives for all software elements. The decision for the HiGHS Solver (CLAIM-005) shows some consideration of technologies and alternatives.
Manage the design	EvidenceStatus.PARTIAL_EVIDENCE	The lack of a formal stakeholder-feedback review (CLAIM-003, CLAIM-007) and architecture-governance process (CLAIM-004) suggests incomplete management regarding capturing design rationale and traceability. However, current architectural implementations (CLAIM-008) indicate some level of design management ensuring adherence to initial design decisions.